Terencio **Agozzino**

Data Scientist · Cloud Engineer · Software Engineer

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rememberYou.github.io | rememberYou | Driver's Licence

Summary

Graduated from an engineering school. Specialized in Cloud and Machine Learning, particularly in Semantic Web and Natural Language Processing (NLP). Through the work done by the Master's thesis at IDLab, the prestigious research center of imec, I learned state-of-the-art techniques in these fields. I co-authored two research papers following my implementation of the `pyRDF2Vec` library, which has over twenty citations to date and is used by many researchers worldwide. Finally, I am mainly passionate about chess, piano, cycling, and writing.

Education

Haute École en Hainaut (HEH) — *Mons, Belgium*

Master, Computer & Systems Engineering — *September 2018 - June 2021*

Learning of engineering skills, including robust database design, a thorough understanding of an operating system, and engineering languages such as C and C++. This engineering curriculum also covers a wide range of skills in automation, electronics, and Machine Learning.

Haute École en Hainaut (HEH) — *Mons, Belgium*

Bachelor, Networks and Telecommunications — *September 2015 - June 2018*

Learning about networks and telecommunications with the current standards offered by Cisco systems.

Université de Mons (UMons) — *Mons, Belgium*

Bachelor, Computer Science — *September 2013 - June 2015*

Learning mathematical logic and refreshing basic knowledge in mathematics and physics. In addition, more in-depth knowledge of computer science is acquired, which allows for better software development.

Certifications

2022	**Professional Scrum Master™ I (PSM I)**, Scrum.org	*827516*
2022	**AWS Certified SysOps Administrator – Associate**, Amazon Web Services (AWS)	*AWS02487783*
2022	**AWS Certified Developer – Associate**, Amazon Web Services (AWS)	*AWS02487783*
2022	**AWS Certified Cloud Practitioner**, Amazon Web Services (AWS)	*AWS02487783*
2021	**AWS Certified Solutions Architect – Associate**, Amazon Web Services (AWS)	*AWS02487783*
2021	**Microsoft Certified: Azure Fundamentals**, Microsoft	*992366501*
2021	**SCO-VCA**, BeSaCC-VCA	*BE000132FR205386*

Publications

pyRDF2Vec: A Python Implementation and Extension of RDF2Vec — *arXiv Preprint*

Gilles Vandewiele, Bram Steenwinckel, **Terencio Agozzino** & Femke Ongenae — *May 2022*

Publication URL: `https://arxiv.org/abs/2205.02283`

INK: Knowledge Graph Embeddings for Node Classification — *Data Mining & Knowledge Discovery*

Bram Steenwinckel, Gilles Vandewiele, Michael Weyns, **Terencio Agozzino**, Filip De Turck & Femke Ongenae — *January 2022*

Publication URL: `https://link.springer.com/article/10.1007/s10618-021-00806-z`

Skills

Fundamentals: AWS | Business Intelligence | CI/CD/CT | Computer Science | Cybersecurity | Data Structure | Knowledge Graphs | Linux | Machine Learning | Microsoft Azure | Natural Language Processing | Networking | REST APIs | Semantic Web | Scrum

Coding: Android | Angular | Bash | C | C# | C++ | Docker | Emacs Lisp | Flutter | GNU Emacs | Git | Java | JavaScript | LaTeX | MongoDB | Node.js | OpenGL | PHP | Python | PyTorch | React | Scikit-learn | TensorFlow | TypeScript | UML | Vue.js

Languages: **French:** Mother Tongue | **English:** Fluent | **Dutch:** Basic | **Italian:** Basic

Experience

Datavillage
Liège, Belgium

Data Scientist — *January 2022 - April 2022*

- Assessed Knowledge Graphs based on metrics (e.g., purity rate, deployment time, usage, and cost).
- Performed a Knowledge Graph migration to provide a more sophisticated recommendation system to end users.
- Improved the recommendation system performance by reducing HTTP latency related to SPARQL queries as well as reducing the algorithmic complexity of Cypher queries and Python implementations.
- Refactored an existing code base in accordance with good architectural practices and design patterns.
- Suggested paths of improvement on the implemented AWS Cloud architecture.
- Used Python, Jupyter Notebook, Neo4j, DBpedia, Wikidata, and Git.

IDLab – imec
Ghent, Belgium

Machine Learning Intern — *February 2021 - June 2021*

- Built a state-of-the-art API, named `pyRDF2Vec`, to facilitate development by the scientific community, choosing appropriate architecture and design patterns as well as implementing linters, Continuous Integration (CI), Continuous Delivery (CD), and Continuous Testing (CT).
- Reduced the processing time of Knowledge Graphs by introducing optimization mechanisms (e.g., use of cache memory to reduce HTTP latencies, multiprocessing, cache prefiling, appropriate data structure of the nodes, and a connection pool).
- Implemented a basic literal support to increase the model accuracy.
- Implemented an online learning solution to prevent an obsolete model from being completely retrained on the Knowledge Graph, but only on the new data added in this Knowledge Graph.
- Implemented and evaluated BERT and other recent embedding techniques within RDF2Vec as part of my Master's thesis.
- Used Python, Jupyter Notebook, Google Colab, LaTeX, Mattermost, and Git.

Open Summer of Code
Brussels, Belgium

Developer (Student Job) — *July 2020 - August 2020*

- Developped a web application which use `RML.io` to generate high quality Linked Data to create Knowledge Graphs.
- Managed a student team to carry out the project.
- Applied the Scrum framework to use the Agile methodology.
- Used React, Node.js, JavaScript, Docker, Slack, and Git.

Open Summer of Code
Brussels, Belgium

Developer (Student Job) — *July 2018 - August 2018*

- Built the web application of Prisma, which has now become `Soulcenter`.
- Performed a project analysis to meet the needs of the end users.
- Applied the Scrum framework to use Agile methodology.
- Used JavaScript, Vue.js, Laravel, Slack, and Git.

Centre de Recherche de la Haute École en Hainaut (CReHEH)
Mons, Belgium

Trainee Developer — *February 2018 - May 2018*

- Designed an API and a CLI for the Haute École en Hainaut as part of my Bachelor's thesis.
- Improved visual recognition and motions of the NAO robot through image processing.
- Built interaction scenarios with the NAO robot to interact with vulnerable children (e.g., with autistic and behavioral disorders).
- Used Python, OpenCV, and Git.

Honors & Awards

2021	**Honorable Mention,** 7th Cyber Security Challenge Belgium	*Brussels, Belgium*
2020	**Honorable Mention,** 6th Cyber Security Challenge Belgium	*Brussels, Belgium*
2020	**Honorable Mention,** 3rd WIRE.HACK Hackaton	*Kortrijk, Belgium*

Presentations

3rd Edition of the Cybersecurity Day by the Haute École en Hainaut
Mons, Belgium

Presenter for <BERT is All You Need> — *December 2020*

- Introduced RNN, the Transformer architecture, and BERT.
- Compared the detection of spam SMS with BERT and other classification algorithms through two Kaggle datasets.

2nd Edition of the Cybersecurity Day by the Haute École en Hainaut
Mons, Belgium

Presenter for <New Vulnerabilities in 4G/5G Networks> — *December 2019*

- Raised public awareness of digital security.
- Introduced User Equipment (UE), Evolve NodeB (eNodeB), and Mobility Management Entity (MME) device types.
- Explained the Long Term Evolution (LTE) registration operation.
- Introduced identification, Bidding-Down, and Depletion-of-Battery attacks.

References

Prof. Dr. Femke Ongenae — *Master's Thesis Advisor*
Assistant Professor · Ghent University — *femke.ongenae@ugent.be*
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Dr. Ir. Gilles Vandewiele — *Master's Thesis Supervisor*
Chief Technology Officer · Optioryx — *gilles.vandewiele@optioryx.com*
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